

# Filip Davchev

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## PROFESSIONAL SUMMARY

I am a Computer Science graduate currently interning as an engineer shipping REST APIs in C#/ASP.NET inside a live production codebase serving web, mobile, and PWA clients. Outside of work I build AI systems end to end: a full RAG pipeline with LangChain and FAISS, a conversational interface with runtime switching across three LLM backends, and prompt engineering with few-shot control over answer grounding. Comfortable across the stack — Python/FastAPI, C#/ASP.NET, JavaScript, Android — and looking to bring that RAG and LLM-orchestration experience into a data infrastructure role, with room to grow into cloud/IaC tooling.

## EXPERIENCE

### Engineer Intern · Imbrium Tech, Shtip — on-site

Aug 2026 — Present

C#/.NET · ASP.NET · REST APIs · SQL Server · Flutter · Azure DevOps · Git

- Built REST API endpoints in ASP.NET serving real-time shipment tracking to three clients — web dashboard, Flutter app, and PWA — in a live production codebase.
- Optimized SQL Server queries and indexing to hold response times steady as shipment volume scaled.

## PROJECTS

### DocuMind AI — Conversational Document Q&A Assistant · Solo Personal Project

2026

Python · LangChain · RAG · FAISS · Embeddings · Streamlit · Ollama · Llama3 · Mistral · LLaVA · pdfplumber

- Built a complete RAG pipeline from scratch — chunking, embedding via nomic-embed-text, FAISS vector search, LangChain orchestration — enabling multi-turn semantic Q&A across multi-page PDFs.
- Designed the prompt layer using in-context (few-shot) prompting to control answer grounding and reduce hallucination on retrieved chunks.
- Shipped the chat UI/UX in Streamlit: conversational input, vision input via LLaVA, runtime switching across 3 LLM backends, and chat history export.

### Classroom Presence System — Contactless Attendance Platform · Group Project

2026

Python · FastAPI · REST APIs · JWT · Firebase Auth · Firestore · Kotlin · Android NFC/HCE · JavaScript

- Built the FastAPI backend end to end — JWT authentication via Firebase Auth, REST endpoints for attendance recording, and Firestore as the live data store.
- Replaced manual roll-call with NFC tap-to-check-in: students tap their phone on the teacher's device and attendance writes to Firestore in real time.

### Book Scanner — Offline Book Cataloguing PWA · Solo Personal Project

2026

TypeScript · React · Vite · Tesseract.js OCR · Gemini API · IndexedDB · Playwright · PWA

- Built an installable PWA that OCR-scans book covers entirely on-device with no server dependency by default, optionally calling the Gemini API for enrichment when online, reaching 15/15 exact title matches.
- Backed the app with an automated test suite — 136 unit tests and 40 end-to-end tests via Playwright.

## TECHNICAL SKILLS

**AI / LLM:** RAG pipelines, LangChain, FAISS, embeddings, prompt engineering, in-context (few-shot) learning, Ollama, Gemini API, Claude Code (AI-assisted development)

**Languages:** Python, C#, Java, Kotlin, JavaScript, SQL, C/C++

**Backend & APIs:** C#/ASP.NET, FastAPI, REST APIs, JWT Authentication, Firebase Admin SDK & Firestore

**Frontend / Mobile:** React, Streamlit, Android (Kotlin/Java), TypeScript

**Databases:** SQL Server, MySQL, Firebase Firestore, Firebase Realtime DB

**Testing & Quality:** Unit testing, Playwright (E2E)

**DevOps / Cloud:** Git, GitHub (feature branches, pull requests, code review), Azure DevOps, Docker (learning), Linux

## EDUCATION

### Bachelor of Science in Computer Science

Fall 2026

University "Goce Delchev" — Shtip, North Macedonia

Relevant coursework: Artificial Intelligence, Machine Learning, Object-Oriented Programming, Data Structures & Algorithms, Database Systems, Software Engineering, Computer Networks.

## INTERESTS

Building small, convenient projects like the book scanner above; also photography, hiking, and basketball.